ClipSAT : Differential Equations First/Second Order, Laplace, Systems & S [ANSWER KEY]

Unit 7.01: Systems of Linear ODEs & Phase Plane Topology

Q1. The matrix of mathbf x'=Amathbf x has eigenvalues pm i . The origin is a:

[ ] A: saddle point unstable
[ ] B: unstable spiral
[ ] C: stable node
[x] D: center stable, not asymptotically

Solution: Complex eigenvalues give spirals; the sign of the real part decides stability, and a zero real part gives a center. Here: center stable, not asymptotically.

Q2. Solve x'=2 x, y'=- y with x0=-1, y0=3 . Find xln{left2 right} .

[ ] A: - frac{1}{2}
[ ] B: 4
[ ] C: 12
[x] D: -4

Solution: The equations are decoupled: x=- e^{2 t} , y=3 e^{- t} . Hence -4 .

Q3. With x_1=y , x_2=y' , the equation y''+3y=0 becomes mathbf x'=Amathbf x . Find A .

[ ] A: left[begin{matrix}0 & 13 & 0end{matr
[x] B: left[begin{matrix}0 & 1-3 & 0end{mat
[ ] C: left[begin{matrix}1 & 0-3 & 0end{mat
[ ] D: left[begin{matrix}0 & 10 & -3end{mat

Solution: x_1'=x_2 and x_2'=y''=- 3 x_{1} , so A=left[begin{matrix}0 & 1-3 & 0end{matrix}right] .

Q4. The matrix of mathbf x'=Amathbf x has eigenvalues pm 5i . The origin is a:

[x] A: center stable, not asymptotically
[ ] B: unstable node
[ ] C: saddle point unstable
[ ] D: unstable spiral

Solution: Complex eigenvalues give spirals; the sign of the real part decides stability, and a zero real part gives a center. Here: center stable, not asymptotically.

Q5. Solve x'=- x, y'=- y with x0=-2, y0=4 . Find yln{left2 right} .

[ ] A: -2
[x] B: 2
[ ] C: -1
[ ] D: 4

Solution: The equations are decoupled: x=- 2 e^{- t} , y=4 e^{- t} . Hence 2 .

Q6. With x_1=y , x_2=y' , the equation y''+y'+y=0 becomes mathbf x'=Amathbf x . Find A .

[x] A: left[begin{matrix}0 & 1-1 & -1end{ma
[ ] B: left[begin{matrix}0 & 11 & 1end{matr
[ ] C: left[begin{matrix}1 & 0-1 & -1end{ma
[ ] D: left[begin{matrix}-1 & -11 & 0end{ma

Solution: x_1'=x_2 and x_2'=y''=- x_{1} - x_{2} , so A=left[begin{matrix}0 & 1-1 & -1end{matrix}right] .

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Q7. The matrix of mathbf x'=Amathbf x has eigenvalues -2pm 4i . The origin is a:

[x] A: stable spiral

[ ] B: unstable spiral

[ ] C: unstable node

[ ] D: stable node

Solution: Complex eigenvalues give spirals; the sign of the real part decides stability, and a zero real part gives a center. Here: stable spiral.

Q8. Solve x'=- x, y'=y with x0=4, y0=-1 . Find x1 .

[ ] A: 4 e

[ ] B: - frac{4}{e}

[ ] C: - frac{1}{e}

[x] D: frac{4}{e}

Solution: The equations are decoupled: x=4 e^{- t} , y=- e^{t} . Hence frac{4}{e} .